## Information for Candidates

### Switch-Mode Power Supplies

Voltage ratio equations for buck SMPS $\frac{V_o}{V_i} = \delta$ or $\frac{V_O}{V_I} = \frac{1}{1+\frac{2fLI_O}{V_I\delta^2}}$

Voltage ratio equations for boost SMPS $\frac{V_o}{V_i} = \frac{1}{1-\delta}$ or $\frac{V_o}{V_i} = \frac{1}{1-\frac{V_i\delta^2}{2fLI_i}}$

### Three-Phase Systems

Line Voltages and Current

Star $V_L = \sqrt{3}\,V_P \quad I_L = I_P$ Delta $V_L = V_P \quad I_L = \sqrt{3}\,I_P$

Power $P_{3\Phi} = 3\,V_P I_P \cos(\phi)$

$= \sqrt{3}\,V_L I_L \cos(\phi)$

### Induction Machines

$$\omega_S = \frac{\omega_E}{P} \qquad s = \frac{\omega_S - \omega_R}{\omega_S} \qquad T_{em} = \frac{3\,I_R'^2\,R_R'}{\omega_R}\left(\frac{1-s}{s}\right)$$

### Photovoltaic Systems

$$I_{PV} = I_{Ph} - I_{AK} - I_{Sh} \qquad I_{AK} = I_0\left[e^{\frac{V_{PV}+I_{PV}R_S}{K_I v_T}} - 1\right]$$

### Power Flow in Lines and Cables

Cable Parameters

$$R'_{LF} = \frac{1}{\sigma_C \pi r_C^2} + \frac{1}{\sigma_C 2\pi r_O t_O} \qquad L' = \frac{\mu_0}{2\pi}\ln\left(r_O/r_C\right) \qquad C' = \frac{2\pi\varepsilon_0\varepsilon_{RI}}{\ln\left(r_O/r_C\right)} \qquad G' = \frac{2\pi\sigma_I}{\ln\left(r_O/r_C\right)}$$

OHL Parameters (approximate form)

$$R'_{LF} = \frac{2}{\sigma_C \pi r_C^2} \qquad L' = \frac{\mu_0}{\pi}\ln\left(d/r_C\right) \qquad C' = \frac{\pi\varepsilon_0\varepsilon_{RI}}{\ln\left(d/r_C\right)} \qquad G' = \frac{\pi\sigma_I}{\ln\left(d/r_C\right)}$$

Power Flow (full form)

$$P_S = \frac{V_S^{\,2}}{Z_{SR}}\cos(\theta) - \frac{V_S V_R}{Z_{SR}}\cos(\theta+\delta)$$

$$Q_S = \frac{V_S^{\,2}}{Z_{SR}}\sin(\theta) - \frac{V_S V_R}{Z_{SR}}\sin(\theta+\delta)$$

Voltage Drop (approximate form)

$$\Delta V = |V_S| - |V_R| \approx \frac{RP_S + XQ_S}{|V_S|}$$

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1. This question covers several topics and all parts should be attempted.

a) Explain why electricity supply systems use three phases with voltages offset by 120°. Explain why three-phase supply is useful in induction machines. [5]

b) Two nodes in a network are connected by a line with an impedance of 0.5 + j1.2 Ω. The voltage at sending end is 33 kV and at receiving end is 32.5 kV with a lag of 1.5º compared to the sending end.

i) Calculate the current in the line and the real and reactive power consumed by the line itself. [4]

ii) Calculate the real and reactive power at the sending end. [3]

c) Consider an AC electricity system operating with a balance between generated power and the consumed power. Explain what happens to the frequency of the system if a large generator is suddenly disconnected. Explain the role of the governors of the remaining generators in controlling the frequency. [7]

d) It is expected that an underground power cable will have a higher capacitance per unit length and a lower inductance per unit length than an equivalent overhead line. Explain why this is expected. [7]

e) Sketch the shape of the current through the inductor, transistor, diode and capacitor for a buck switch-mode power supply (SMPS). Assuming the ripple is small and the average inductor current is 10 A, calculate the conduction loss in the transistor and diode given that the transistor has an on-state resistance of 25 mΩ and the diode has an on-state voltage of 0.7 V and the SMPS is running with a duty cycle of 0.55. [7]

f) Describe how a maximum power point tracking algorithm for a PV panel works and why it is needed. Explain what factors may cause the output from a PV panel to vary during the course of a single day. [7]

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2.

a) The rotor of an induction machine has no electrical connection to an external supply. Explain why this is a desirable feature and also explain how it is that torque is exerted on the rotor. [6]

b) A 3-phase, 2 pole-pair, three-phase induction machine is star-connected and provided with a supply with a phase voltage of 2,000 V, 50 Hz. The parameters of the machine's equivalent circuit, referred to the stator, are:

| | |
|---|---|
| stator resistance | 5.0 Ω, |
| stator leakage reactance | 15.0 Ω, |
| magnetising resistance | 8,000 Ω. |
| magnetising reactance | 900 Ω, |
| referred rotor resistance | 3.5 Ω, |
| referred rotor leakage reactance | 8.0 Ω, |

When driving a particular mechanical load, the machine is observed to spin at 1,455 rpm.

i) Calculate the slip. [3]

ii) Give an expression for the total input impedance of the machine and calculate its value in complex form. [7]

iii) Calculate the electro-magnetic torque. [6]

iv) Calculate the efficiency. [4]

c) Explain what is expected to happen to the rotor current and torque if the slip is doubled. There is no need to calculate numerical answers but do state any approximations used in the argument. [4]

3.

A **boost** SMPS has the following design properties.

| | |
|---|---|
| Input voltage | $V_I = 6.0$ V |
| Maximum output current | $I_O^{max} = 0.5$ A |
| Switching frequency | $f = 100$ kHz |
| Inductor | $L = 500$ μH |
| Capacitor | $C = 1{,}000$ μF with ESR = 25 mΩ |

a) Consider the boost SMPS operating in continuous mode.

i) Calculate the duty-cycle at which the circuit should be operated to achieve an output voltage of $V_O = 18.0$ V. [3]

ii) Calculate the ripple component of the inductor current. [3]

iii) Find the average value of the current through the diode and the average current through the inductor when the load draws the maximum output current. [4]

iv) Calculate the voltage ripple across the capacitor when operating at the maximum output current. [8]

v) Calculate the value of inductor current at which the SMPS enters discontinuous conduction mode and the value of output current to which this corresponds. [4]

b) Consider now the boost SMPS operating in discontinuous mode with a duty cycle of 0.5.

i) Calculate the energy stored in the inductor at the end of the on-time. [4]

ii) Find the value of output voltage that the SMPS will adopt under these conditions if the output is connected to a load resistor of a 600 Ω. [4]